

Matias Barandiaran

(+49) 152 3162 6943 • mbarandiaranr@gmail.com • mbaranr.github.io

Education

Lancaster University

Bachelor of Science in Computer Science

Leipzig, Germany

October 2022 – July 2025

- Ranked **1st** out of **212** students.
- First Class Honours (**UK**: 91.2% $\approx$ **GPA**: 4.0/4.0).
- University ranked **#132** worldwide for Computer Science (QS by subject) in 2025.

San Ignacio de Recalde School

International Baccalaureate

Lima, Peru

March 2020 – December 2022

- Cohort's top 5% (**GPA**: 3.9/4.0).

Academic Gaps

August 2025 – Present: Prepared Master's application, presented at International Conference on Artificial Life 2025, enrolled in German language lessons.

January 2022 – September 2022: Transition period between end of Peruvian school year and start of UK undergraduate year. Completed visa processing, relocation, and academic preparation.

Work Experience

Technical University of Munich

Research Assistant (Hiwi)

Munich, Germany

July 2024 – August 2025

- Hosted by the Gagneur Lab and Helmholtz Munich.
- Worked on rare variant association testing using deep learning and data-driven burden scores (DeepRVAT), utilizing **Python**, **R**, **PyTorch**, and **Snakemake**.

Martin Luther University of Halle-Wittenberg

Research Intern

Remote

July 2024 – September 2024

- Hosted by the Faculty of Medicine.
- Worked on novel graph-level anomaly detection via latent space searching and generative adversarial networks (gladGAN) on molecular datasets, utilizing **Python** and **Pytorch**, and **scikit-learn**.

Lancaster University Leipzig

Teaching Assistant

Leipzig, Germany

October 2023 – June 2024

- Laboratory assistant and demonstrator for the Digital Systems module.
- Delivered instructional sessions of low-level **MIPS** assembly programming.

Charles University

Research Intern

Prague, Czechia

August 2023 – September 2023

- Hosted by the Faculty of Mathematics and Physics.

- Contributed to: *Unsatisfiability Proofs for Horn Solving* (2025). *ETAPS best paper award*
- Designed production of Alethe verifiable proofs for the Golem horn clause solver in C++.

Freelance

Lima, Peru

Math Tutor

August 2020 – September 2021

- Guided senior high school students to achieve outstanding exam performance and substantial growth in their mathematical abilities (IB Mathematics SL and HL).

Awards / Honours

Chancellor's Medal – Lancaster University (2025): The university's most prestigious award, presented annually to graduating students who demonstrate exceptional merit in their field.

Valedictorian – Lancaster University (2025): Ranked **1st** out of **212** students achieving the cohort's highest average grade across all three years of the BSc Computer Science program.

Outstanding Dissertation – Lancaster University (2025): Obtained the highest achievable grade (**100%**) for the BSc Computer Science dissertation: *Growing Reservoirs with Developmental Graph Cellular Automata*. **Strong Accept** for presentation at peer-reviewed International Conference on Artificial Life 2025 in Kyoto, Japan.

EASST Best Paper Award – Charles University (2025): Best ETAPS (International Joint Conferences On Theory and Practice of Software) 2025 paper related to the systematic and rigorous engineering of software and systems.

Scholarship – Lancaster University (2022): Awarded a €4,000 tuition scholarship in recognition of academic merit.

Leadership Activities

Lancaster University

Leipzig, Germany

Founder, Computer Science Society

2023 – 2025

- Dedicated to fostering a culture of innovation, diversity and problem-solving.
- Secured university funding for the purchase of robotics supplies (e.g., 3D printer, Arduinos, etc.).

Lancaster University

Bailrigg, England

Co-Organizer, Global Advancement Fund

2024

- Co-Initiated and championed the approval process of a ~€20k fund to facilitate international collaboration between Lancaster campuses worldwide.

Lancaster University

Leipzig, Germany

Lead Organizer, Qiskit Fall Fest

2023

- Quantum Computing (QC) event powered by IBM.
- Gave an introductory talk on QC and held workshops with Qiskit material. Assembled a series of seminars with special guests like Enrique Solano (Co-CEO of Kipu Quantum).

Publications

Matias Barandiaran and James Stovold. Growing reservoirs with developmental graph cellular automata. arXiv preprint arXiv:2508.08091, 2025.

Rodrigo Otoni, Martin Blicha, Matias Barandiaran, Patrick Eugster, Jan Kofrön, and Natasha Sharygina. Unsatisfiability proofs for horn solving. In the International Conference on Tools and Algorithms for the Construction and Analysis of Systems, pages 67–87. Springer, 2025.

Projects

Industrial Waste Sorting Arm – Lancaster University (2024): Integrated an industrial robotic arm with object detection and image classification software for the purpose of automated waste sorting, utilizing **Python**, **PyTorch**, **C++**, **C**, and **OpenCV**. Funded by Lancaster University’s Global Advancement Fund.

Global Game Jam – Lancaster University (2023, 2024): Participated in 48-hour game development events using **Java** and **libGDX**. Collaborated in a team using **agile development** and **continuous integration** practices.

Certifications

Advanced Applications of Neural Networks and Deep Learning – University of Oxford (2023): Worked with Generative Adversarial Networks, Variational Autoencoders, Deep Reinforcement Learning, Diffusion Models and Graph Neural Networks. Overall grade: A+.

Hardware of a Quantum Computer – Delft University of Technology (2023): Studied the building blocks of quantum computers. Learned how quantum gates are implemented across qubit types.

Additional Skills and Interests

Languages: Spanish (native), English (C2) ***TOEFL**: 114/120*, German (A2).

Tools: Python, Java, R, PyTorch, Qiskit, Git, C, C++, SQL, Linux, Snakemake, Qiskit, QEMU, OS-Ken.

Concepts: Rare Variant Association Testing, Data Pipelines, Computer Vision, Generative Modelling, Supervised Learning, Reinforcement Learning, Reservoir Computing, Cloud Computing.

Hobbies: Game Development, Scuba Diving, Badminton.